

Data Handling: Import, Cleaning and Visualisation

Exercise 5: Data Preparation and Data Analysis with R

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Exercises¹

The GitHub repo for Exercise Session 5 contains csv files with data needed to solve the following exercises. Make sure to store them in your `r_course/data/` folder. Do not forget to load `tidyverse` before you get started!

Exercise A

Read `stocks.csv` into RStudio (store it in variable `stocks`). The dataset is currently in long format. Use `spread()` to transform it into wide format (separate columns for different years). Store the resulting tibble/data.frame in a variable called `stocks2`. Then, run the following code.

```
gather(stocks2, "year", "return", `2015`:`2016`) == stocks
```

Why is the result not identical to the original dataset (`stocks`)? In other words, why are `gather()` and `spread()` not perfectly symmetrical? (Hint: look at the variable types and think about column names.)

Exercise B

Read `countries.csv` into RStudio (store it in variable `countries`). Then, run the following code.

```
gather(countries, 1999, 2000, key = "year", value = "cases")
```

Why does this code fail? How can you fix the problem?

Exercise C

Read `people.csv` into RStudio (store it in variable `people`). Try to spread `people`. Why does the spreading fail? How could you add a new column to fix the problem?

Exercise D

Read `preg.csv` into RStudio (store it in variable `preg`). Tidy `preg`. Do you need to spread or gather it? What are the variables?

Additional Exercises (advanced)

Exercise E

Load `covid_sample.csv` (this is a small sample of the data from the EU Open Data Portal that we have seen in previous exercise sessions). Spread the data (i.e., you want one row for each country and columns of the

¹Adopted from <https://r4ds.had.co.nz/tidy-data.html>.

kind deaths_20/09/2020 and cases_20/09/2020). Hint: check out `pivot_longer()` and `pivot_wider()` (from `tidyr`) – which one of the two could be helpful?

Mock Exam Questions

Part 1: TRUE/FALSE Questions

For each of the following statements, decide whether the statement is TRUE (T) or FALSE (F).

1. The following line of R code will return "bnn".

```
str_split("banana," pattern= "a")
```

Part 2: Multiple Choice Type A

For each of the following questions/statements, decide which of the possible responses are correct. NOTE: Either ONE OR SEVERAL of the response statements can be correct. In order to get a point, all correct response statements need to be marked!

1. Regarding the data semantics (vocabulary used to describe data) introduced in this course, which of the following statements is correct?

- A) A variable contains all values that measure the same underlying attribute across units/observations.
 - B) Datasets typically contain more variables than values.
 - C) An observation contains all values measured on the same unit (e.g., a person).
 - D) Every observation belongs to a value and a variable.
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Part 3: Multiple Choice Type B (only one correct!)

For each of the following questions/statements, decide which of the possible responses is correct. NOTE: For each question only ONE of the possible responses is correct. Select ONLY ONE of the response statements per question!

1. Which of the following statements about 'tidy' data is correct?

- A) When a csv-file is imported with `read_csv()`, the function issues a warning if the data imported is not tidy.
- B) `tibbles` can only contain tidy data.
- C) Data stored in csv-format is generally referred to as 'tidy data'.
- D) Datasets in wide format are generally not considered 'tidy'.